**Scenario**

A security team wants to rapidly quarantine a compromised host by isolating it from the rest of the network.

**Question:** Describe how an SDN controller, leveraging OpenFlow, can achieve this rapid isolation without reconfiguring physical ports or VLANs.

**Answer:**

An SDN controller leveraging OpenFlow can rapidly quarantine a compromised host dynamically in software across all switches without needing physical cable changes, port reconfigurations, or VLAN provisioning through the following flow installation process:

1. **High-Priority Drop Flow Rule Generation:**
   Upon receiving an alert from a Security Information and Event Management (SIEM) system or Intrusion Prevention System (IPS), the SDN controller instantly constructs high-priority OpenFlow `OFPT_FLOW_MOD` messages specifying isolation rules for the compromised host's IP and/or MAC address.
2. **Targeted Match Fields & Explicit Drop Actions:**
   The controller generates flow entries containing:
   - **Match Criteria:** Matches packets where the Source IP/MAC address equals the compromised host (e.g., `eth_src=MAC_COMPROMISED` or `ipv4_src=IP_COMPROMISED`) as well as Destination IP/MAC address for incoming traffic to that host.
   - **Priority Level:** Sets a high priority value (e.g., `priority=65535`, the maximum OpenFlow priority) to ensure this entry takes precedence over any existing forwarding or routing flow rules in the flow tables.
   - **Action:** Sets the action list to empty or explicitly specifies `Action: DROP`.
3. **Proactive Flow-Mod Injection Across Network Switches:**
   The controller pushes these high-priority `OFPT_FLOW_MOD` commands via the southbound interface directly to the ingress switch port where the host resides (or globally across all switch flow tables in the network).
4. **Immediate Hardware Line-Rate Quarantine:**
   The switches immediately update their Ternary Content-Addressable Memory (TCAM) flow tables. Any subsequent packet sent by or addressed to the compromised host instantly hits the high-priority quarantine entry and is dropped at line rate in hardware.
5. **Optional Dynamic Redirection to Remediation Server:**
   Alternatively, instead of dropping all packets, the controller can push a `Flow-Mod` that rewrites header fields (MAC/IP) and redirects traffic exclusively to an isolated sandbox or remediation

server for forensic inspection, completely cutting off access to internal corporate systems and external networks.